Alex Santora

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Education

Stevens Institute of Technology Hoboken, NJ

B.E. Computer Engineering; Applied Mathematics *Sep. 2025 - May 2028*

- **GPA: 3.88/4.00**; Dean's List; Edwin A. Stevens Scholarship; NCAA varsity baseball student-athlete.
- **Coursework:** Data Structures and Algorithms, Microprocessor Systems, Multi-variable Calculus, Linear Algebra

Experience

Stevens Institute of Technology Hoboken, NJ

Undergraduate Research Intern *Sept. 2026 – Present*

- Developing an **AI text provenance benchmark** to evaluate how reliably generated content can be traced across **Qwen 8B/32B, DeepSeek Flash, and Llama 4.**
- Testing provenance across **light polishing, complete AI rewrites, and human-edited text** to measure how repeated rewriting affects model attribution, provenance retention, and AI-detection accuracy.

University of New Hampshire Durham, NH

Undergraduate Research Intern *May 2026 – Aug. 2026*

- **Co-authored an accepted research paper** on MDD-LPC, a privacy-preserving speech anonymization method, for the **32nd Asia and South Pacific Design Automation Conference (ASP-DAC 2027)**.
- Developed and evaluated MDD-LPC, achieving **33.6× faster processing** and **20.24% lower power consumption** than the GenGAN deep-learning baseline while reducing latency from **17.50s to 0.521s**.
- Evaluated privacy–utility tradeoffs across **SVM, CNN, GRU-Mixer, and Wav2Vec2** using CREMA-D, RAVDESS, and TAME Pain, reducing gender-inference F1 by **23.9–66.8%** while preserving emotion and pain-detection performance.

Projects

Baseball Pitch Classification Model | *Personal Project* Jun. 2026

- Achieved **93.98%** pitch-type classification accuracy and **0.94 weighted F1** on an **87,113-pitch** test set by training a balanced Random Forest on **350K+** Statcast-style pitch records and engineering features from velocity, spin, movement, release point, and extension.
- Built a reproducible **end-to-end ML pipeline** for automated data ingestion, preprocessing, training, model persistence, and evaluation using pandas, scikit-learn, joblib, and pybaseball.

AI Travel Planning & Optimization Agent | *Python, LLMs, APIs, Optimization* Oct. 2026

- Built an **end-to-end AI travel agent** integrating **4 external tools** for route planning, web search, weather forecasting, and budget optimization, generating complete itineraries across flights, ground transit, lodging, and activities.
- Targeted **96% itinerary feasibility** across **100 test scenarios** and **18% lower total travel costs** versus a flight-first baseline through autonomous constraint validation, multi-modal optimization, and iterative replanning.

Leadership

NCAA Varsity Baseball Stevens Institute of Technology

Student-Athlete; Leadership Council Member *Sep. 2025 – Present*

- Serve on the **Stevens Baseball Leadership Council**, promoting team accountability, communication, and a strong team-first culture while competing as an **NCAA varsity student-athlete**.

FutureWithAI.dev

Founder & Developer *Mar. 2026 – Present*

- Founded an **AI-literacy and mentorship platform** reaching **500+ students** and growing to **5K+ followers** with **1M+ views**, while leading school presentations and mentoring students in AI, machine learning, and technical projects.

Technical Skills

Languages: Python, C++, Java, JavaScript, Arduino
Tools & Technologies: Git, SolidWorks, Pandas, PyTorch, scikit-learn, NumPy